ONE-PAGE BRIEF

# Post-quantum cryptography review: besu @ 1f1ebc2573e1

Client: Hyperledger Besu — public repository (passive review) · Source: github.com/besu-eth/besu, commit `1f1ebc2573e1` · Reviewed: 2026-11-06 15:36 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

**Level 0 of 4**

no post-quantum cryptography

FINDINGS

**0** Critical **0** High **7** Medium

**0** Low **0** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Key establishment: 10 + 5 − 8 = 7 → **too late**

Signatures: 10 + 5 − 8 = 7 → **too late**

**Next step:** Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

## WHAT MATTERS MOST

| | | |
|---|---|---|
| Medium | QSI-2026-19546 | **Quantum-vulnerable ECDH: key wrapping and data encryption at rest (1 file(s))** preliminary<br>What to do: Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may already have been copied needs its... |
| Medium | QSI-2026-19543 | **Quantum-vulnerable ECDSA: transaction and message signing (6 file(s))** preliminary<br>What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains... |
| Medium | QSI-2026-19544 | **Quantum-vulnerable BLS12-381: transaction and message signing (11 file(s))** preliminary<br>What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains... |
| Medium | QSI-2026-19545 | **Quantum-vulnerable RSA: transaction and message signing (1 file(s))** preliminary<br>What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains... |

3 more in the full report.

## WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

| Role | Algorithms | What to do |
|---|---|---|
| Medium key wrapping and data encryption at rest | ECDH | Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may... |
| Medium transaction and message signing | BLS12-381, ECDSA, RSA | Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum... |
| Medium role not determined | ECDSA, ECDH, BN254 (alt_bn128) | Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)... |

Static review of source code and manifests; files analysed: 2927. No requests were sent to the owner's systems.

Full report, content SHA-256: `25c60fc9a59edc6c0ea69ca2871ea1967c583c30f639e09b309380192ecf324d`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.